Experiment Title: Design and Implementation of ATM System Use Case Diagram

Aim:

To design a UML Use Case Diagram for an ATM System using UML tool to represent the interaction between users and the system.

Introduction:

An Automated Teller Machine (ATM) allows customers to perform banking transactions such as withdrawal, deposit, and balance inquiry. UML (Unified Modeling Language) is used to visually represent system requirements and behavior. The Use Case Diagram models the functional interaction between actors and the system.

Problem Description:

Develop a UML Use Case Diagram for an ATM System that includes authentication, transaction processing, receipt generation, and card ejection. The system must interact with a Bank Server for validation.

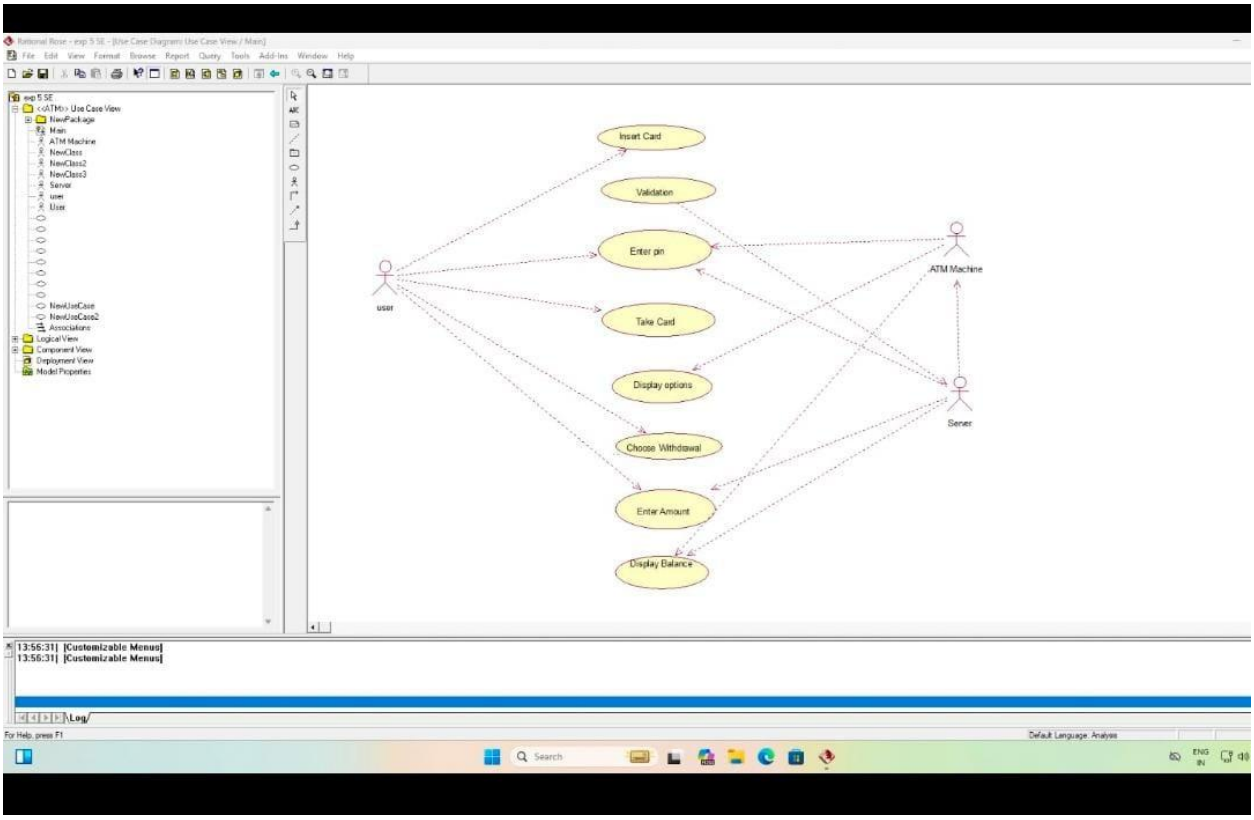
Objective:

- To understand UML Use Case Diagram concepts
- To identify actors and use cases
- To represent system boundary
- To model include relationships
- To design diagram using StarUML

Design Process:

1. Open StarUML and create a new UML project.
2. Add a Use Case Diagram.
3. Create actors: Customer and Bank Server.
4. Draw system boundary labeled "ATM System".
5. Add use cases inside the boundary.
6. Connect actors using association lines.
7. Add <<include>> relationships for mandatory steps.
8. Arrange diagram neatly.

Use Case Diagram:



System Overview

The ATM system authenticates users and allows them to perform banking transactions. It communicates with the Bank Server for validation and ensures receipt generation and card ejection after transaction completion.

Actors Identified:

- **Customer** – Primary actor who performs transactions.
- **Bank Server** – Secondary actor that validates transactions.

Use Cases Identified:

- Authenticate User
- Withdraw Cash

- Deposit Money
- Check Balance
- Print Receipt
- Eject Card

Functional Requirements:

- System must verify user credentials.
- System must allow withdrawal, deposit, and balance checking.
- System must generate receipt after transaction.
- System must eject card after completion.

Design Constraints:

- Must follow UML standards.
- Actors placed outside system boundary.
- Use cases inside boundary.
- Include relationships shown using dashed arrows.
- Diagram created using StarUML.

Observation:

The Use Case Diagram clearly represents system functionality and actor interaction. The <<include>> relationship correctly models mandatory receipt generation. The diagram structure follows proper UML conventions.

Result:

The ATM Use Case Diagram was successfully created in StarUML. All required actors, use cases, and relationships were accurately modeled.

Conclusion:

The experiment successfully demonstrated how UML Use Case Diagrams can be used to model functional requirements of an ATM System. The diagram improves understanding of system behavior and interaction between components.